



HACKTHEBOX

HyperFileTable Writeup



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Prepared by: Blitztide

Machine Author(s): Blitztide

Difficulty: **Easy**

Scenario

There has been a new joiner in Forela, they have downloaded their onboarding documentation, however someone has managed to phish the user with a malicious attachment.

Artefacts Provided

- HyperFileTable.zip - 9070aa868d6f0592c838be2f52cc185320b1a90b

Initial Analysis

To start the Investigation we load the `hyperfiletable.zip` into our forensics workstation, unzip using `hacktheblue` and then expand the contained `mft.raw.bz2` using 7zip.

We can now verify that the MFT (Master File Table) is intact by using the following powershell command.

We can now use the tool `MFTEcmd` to convert the MFT into a CSV for parsing, whilst we wait for `MFTEexplorer` to load the MFT.

```
> MFTEcmd.exe -f .\mft.raw --csv csvout
MFTEcmd version 1.2.2.1

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/MFTEcmd

Command line: -f .\mft.raw --csv csvout

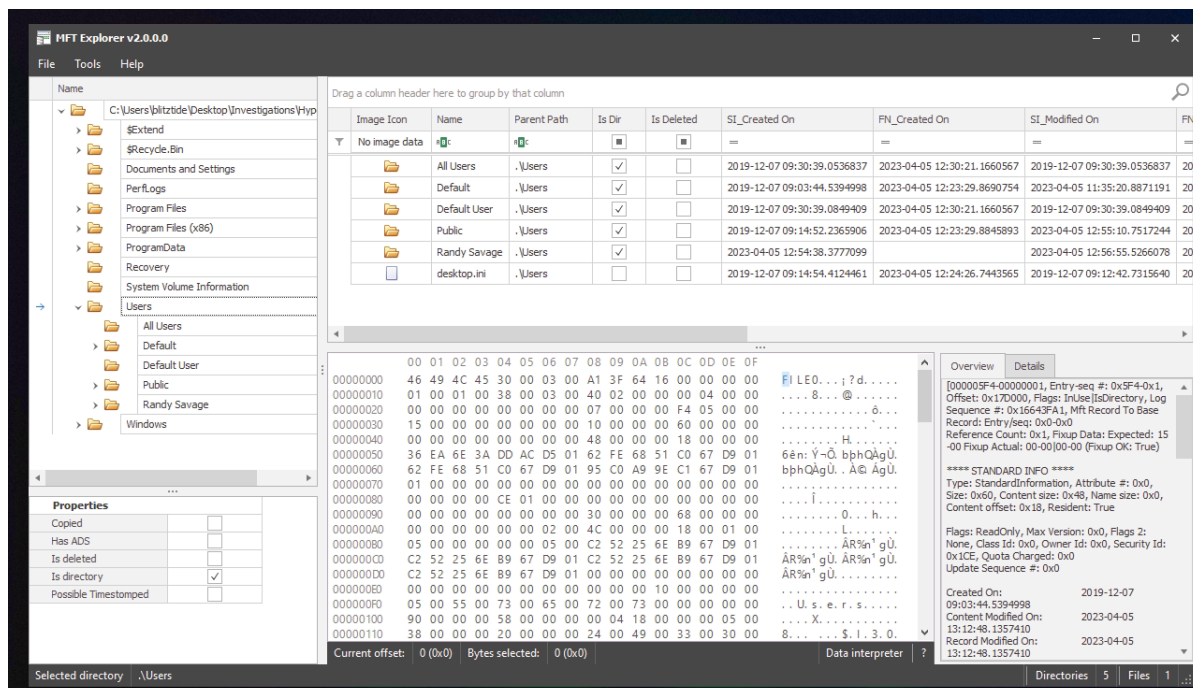
Warning: Administrator privileges not found!

File type: Mft

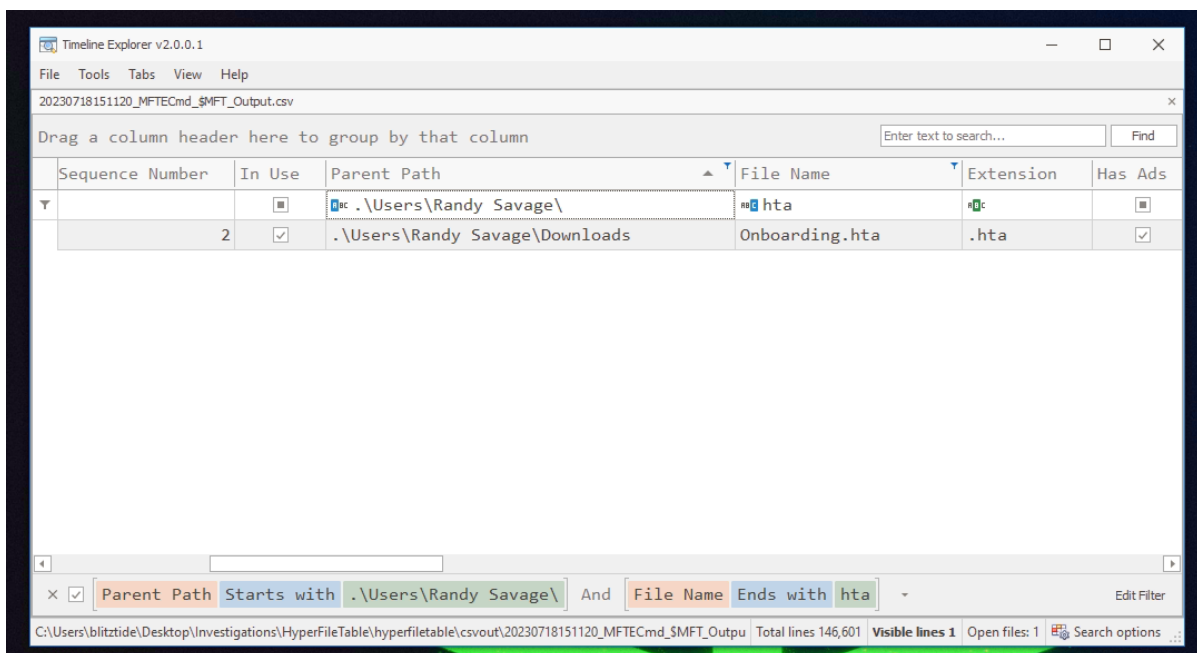
Processed .\mft.raw in 4.0921 seconds

.\mft.raw: FILE records found: 110,818 (Free records: 7,240) File size: 115.5MB
Path to csvout doesn't exist. Creating...
    CSV output will be saved to csvout\20230718151120_MFTEcmd_.$MFT_Output.csv
```

When we load the CSV into `MFTEexplorer` we are able to see that there is only one folder under `C:\Users` that isn't default which is the user `Randy Savage`.

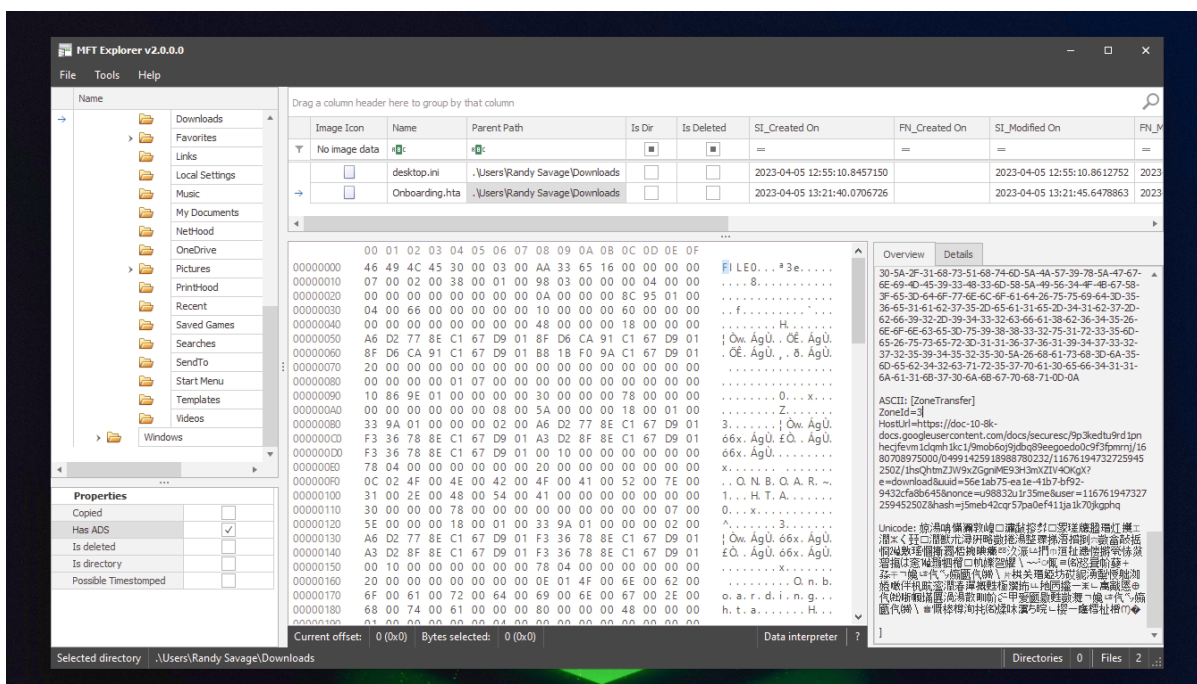


To identify HTA files that have been downloaded by `Randy Savage` we can use search filters in `TimeLineExplorer` looking for files that end in `.hta` and under `C:\Users\Randy Savage`.



The next question refers to a ZoneID and download url for the file, this is telling us to look at **Alternate Data Streams** for the file, a common one known as MOTW (Mark of the Web) is found under the **ADS.Zone.Identifier**.

The **Zone.Identifier** ADS is what Windows uses to track where files have come from, we can see this in **MFTE Explorer**



This shows us the download URL of **https://doc-10-8k-docs.googleusercontent.com/docs/securesc/9p3kedtu9rd1pnhecjfevm1clqmh1kc1/9mob60j9jdbq89eegoedo0c9f3fpmrnj/1680708975000/04991425918988780232/11676194732725945250Z/1hsQht mZJW9xZGgniME93H3mXZIV40KGX?e=download&uuid=56e1ab75-ea1e-41b7-bf92-9432cfa8b645&nonce=u98832u1r35me&user=11676194732725945250Z&hash=j5m eb42cqr57pa0ef411ja1k70jkqphq** and the ZoneID of **3** which indicates that the file has been retrieved from the Internet.

We can also find the allocated size and real size for the HTA file from this entry, the size of the file is **1144** bytes which is under the standard allocation of **4096** bytes, so the allocated size is **4096**.

Browsing around the User's documents we can find a folder `C:\Users\Randy Savage\Work` which contains three deleted files:

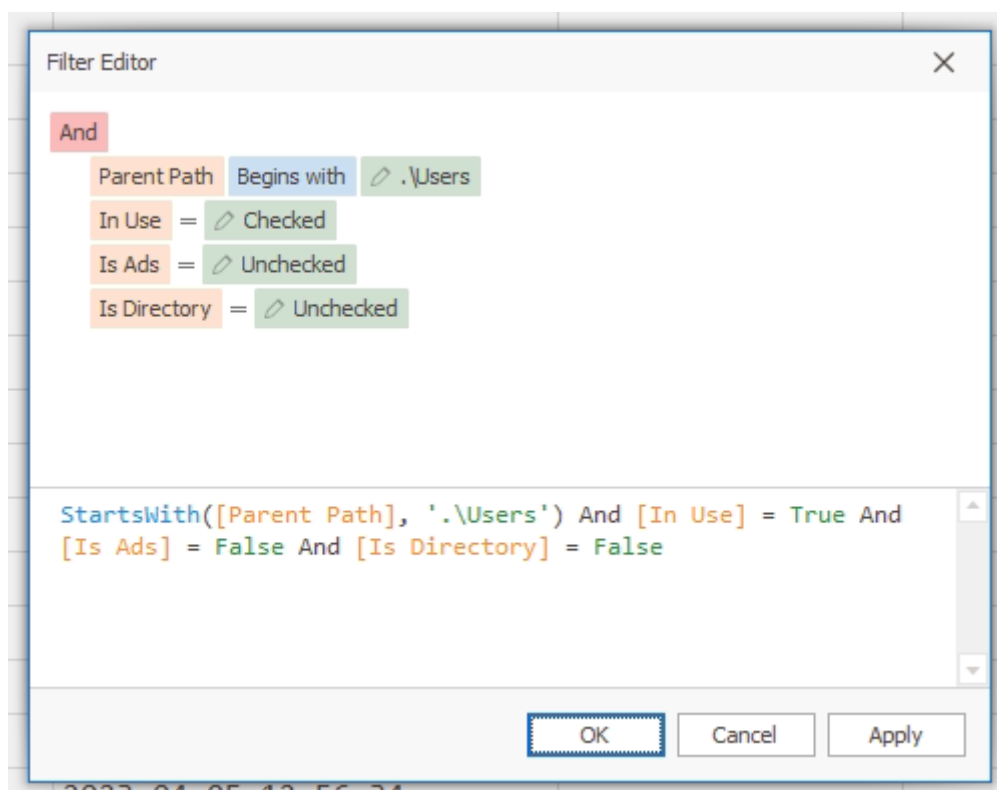
- Notes.txt - MFT Resident - deleted
- Proposal.pptx - Non-Resident - deleted
- Stats.xls - Non-Resident - deleted

We can view the contents of `notes.txt` as the filesize is small enough for its contents to be stored entirely inside an MFT entry.

New onboarding process:
Download onboarding tool from Google Drive
Username: RSavage
Password: ReallyC00lDucks2023!

We can find the time that `Proposal.pptx` was downloaded by using the `Standard Information` Attribute Created On time which returns `2023-04-05 13:11:49.7425214`

The final question is tricky, but we can use Timeline explorer to help us filter all files that we require.



This returns a value of `3471` files remaining on the system.

Questions

1. What is the MD5 hash of the MFT?

`3730c2fedcdc3ecd9b83cbea08373226`

2. What is the name of the only user on the system?

`Randy Savage`

3. What is the name of the malicious HTA that was downloaded by that user?

Onboarding.hta

4. What is the ZoneID of the download for the malicious HTA file?

3

5. What is the download URL for the malicious HTA?

https://doc-10-8k-

docs.googleusercontent.com/docs/securesc/9p3kedtu9rd1pnhecjfevm1clqmh1kc1/9mob6oj9jdbq89eegoedo0c9f3fpmrnj/1680708975000/04991425918988780232/11676194732725945250Z/1hsQhtmZJW9xZGgniME93H3mXZIV40KgX?e=download&uuid=56e1ab75-ea1e-41b7-bf92-9432cfa8b645&nonce=u98832u1r35me&user=11676194732725945250Z&hash=j5meb42cqr57pa0ef411ja1k70jkgphq

6. What is the allocated size for the HTA file?

4096

7. What is the real size of the HTA file?

1144

8. When was the powerpoint presentation downloaded by the user?

2023-04-05 13:11:49

9. The user has made notes of their work credentials, what is their password?

ReallyC00lDucks2023!

10. How many files remain under the C:\Users\ directory? (Recursively)

3471